DIY Jon-Air Airboat: Build Guide

Version 1.0.1.0, minimal 1448 build, Rotax 800R E-TEC. 199 DIY hours, 95 pro hours (estimates). Generated from the Build Guide tab of the tracker; BOM line numbers refer to that tracker.

Section 1 - Hull Armor

1.1 Buy a used 1448 jonboat

Depends on: - | Alongside: - | About 10 h DIY, 0 h pro

1. Raise the purchase money (see BOM line 1 for the current price estimate).
2. Search Facebook Marketplace, Craigslist (Orange County, LA, Inland Empire, Ventura) and OfferUp for "1448", "14 ft jon boat", "jonboat". Set a saved search / alert.
3. Target: 14 ft long, 48 in bottom width, riveted or welded aluminum, flat bottom with ribs inside. Lowe, Tracker, Alumacraft, Crestliner, Polar Kraft are all fine. A trailer is a big plus.
4. Before visiting, ask for: photos of the bottom and transom, title / registration (CA "pink slip") in the seller's name, and the HIN (hull ID number) on the transom.
5. Inspect in person: look for cracks around the transom and ribs, corrosion (white powder, pitting), dents with creases, and rivets with a dark halo (weeping). Ask to put a few inches of water in the hull (or look at it on the water) and check underneath for drips.
6. Negotiate. If the boat comes with a motor you don't need, factor in its resale value.
7. Pay, get the signed title and a bill of sale, and transfer registration at the DMV within 10 days.
8. Tow it home (check that trailer lights and tires work) or arrange transport.

Skills: Fundraising; Use of online 'for sale' forums

1.2 Add UHMW strips to the bottom of the hull

Depends on: 1.1 | Alongside: 2.1, 2.8 (shop work) | About 16 h DIY, 8 h pro

MINIMAL BUILD (reports/Drivetrain_800R_ETEC.md, Section 8): 3/16 in UHMW over the wear zones only, three 12 in strips: one on the centerline and one at each chine, from the transom to about 135 in (about 32 lb). The full 8-strip 1/4 in layout below weighs about 75 lb and does not fit the payload budget; use it only for the strip positions and fastening.

Layout from the Onshape model (Lowe 1448 reference; verify on your hull): 8 strips, 1/4 in UHMW, 1/4 in gaps. Center strips 12 in wide; outer strips run to the chine (about 12 in). Center strips break at the 54.75 in rib, outer strips at the 95.25 in rib (staggered seams). UHMW runs from the transom to about 135 in forward, about 2/3 of the way up the bow rise.

Cut list (from two 4x8 sheets, each ripped into four 12 in strips): 2 @ 95.25 in (outer aft), 2 @ ~40 in (outer fwd), 2 @ 54.75 in (center aft), 2 @ ~81 in (center fwd). Cut fwd pieces 2 in long and trim to fit.

1. Flip the hull upside down onto sawhorses or old tires with a helper (about 150 lb). Never work under a hull that is not blocked securely.
2. Clean the bottom: pressure wash, scrape off growth, degrease with acetone. Leak-test first and fix any weeping rivets now; the UHMW will hide them.
3. From inside the hull, measure each rib's distance from the transom and mark it on the outside (approx. 37.9, 54.75, 70.75, 95.25, 108.25 and 122.25 in on the reference hull). Snap a centerline and mark the 12

in strip lines.

4. Rip the sheets into 12 in strips with a circular saw (fine-tooth plywood blade) against a straightedge; cut to length. Round the corners and chamfer the forward (leading) edges with a file or router.
5. Work at a moderate temperature (60-80 F). Leave 1/4 in between strips and 1/8 in at seams; UHMW grows and shrinks about 10x as much as aluminum.
6. Clamp the first strip in place. At each rib, drill a 3/16 in pilot through UHMW and hull, centered on the rib flange and at least 1/2 in from existing rivets (4 bolts across the bottom per strip pair: 3-1/8 and 9-1/8 in from centerline on center strips, 15-7/8 and 20-7/8 in on outer strips; at seams, one bolt 1 in each side of the seam).
7. Open all holes to #7 (13/64 in). Fixed row = the forward rib of each piece: leave round. All other rows: slot the hole in the UHMW only, about 5/8 in long fore-aft (drill two holes and file between them). The hull hole stays round.
8. Countersink the UHMW with the 82 deg bit until the screw head sits flush to 1/32 in below the surface. Test-fit one screw.
9. Remove the strip, deburr the aluminum holes, wipe with acetone.
10. Reinstall the strip. Put a dab of 4000UV in each hull hole and on each screw under the head. Insert #10-24 x 1 in flat-head screws from outside.
11. Inside the hull, put a fender washer and a dab of sealant on the rib flange, then the nylock nut. A helper holds the screwdriver outside. Tighten the fixed row snug; tighten slotted rows just until the head seats (do not crush the UHMW).
12. Repeat for all 8 pieces, wiping off excess sealant. Let cure 24 h.
13. Flip the hull back over. Water-test (a few inches of water in the hull, or launch) and check every bolt from outside / inside for drips; re-bed any that weep.

Skills: Measuring & layout; Power drill; Circular saw (straight cuts); Countersinking; Applying marine sealant; Basic hand tools (wrench, screwdriver); Safe lifting/flipping with a helper

Section 2 - Install Drivetrain And Prop Cage

MINIMAL BUILD: a Rotax 800R E-TEC snowmobile two-stroke driving the prop through one 8MGT belt, on a bolted aluminum stand, with a smaller aluminum cage. The factory benches stay. The driver sits on the middle bench. Design details are in reports/Drivetrain_800R_ETEC.md; wiring is in electrical/Minimal_1448_Electrical_Schematic.pdf and electrical/Wire_List.csv.

Dimensions use the CAD frame: X forward from the transom, Y + to port, Z up from the hull bottom, all in inches. They come from a reference model of a Lowe 1448. Check them on your own hull, and stop if anything differs by more than 1/2 in.

Approach: build the stand and drivetrain as a module off the boat and lower it onto waiting shoes. Then connect the systems, build the cage around the real prop position, and fit the prop last.

Step images show the v1.0.0.4 layout. Where an image differs from the text, follow the text: the radiator has moved, and the oil tank, battery switch, side lights and floor conduit are new.

2.0 Choose your prop (standard 3-blade, or the quieter 4-blade)

Depends on: - | Alongside: 1.1, 1.2 | About 1 h DIY, 0 h pro

Decide this before building the belt reduction (step 2.8): the prop sets the large pulley size.

STANDARD (this guide and the BOM): Warp Drive 3-blade airboat prop with HP blades, cut to 66 in (BOM line 32, $1,275 plus face plate and flange). Belt 8MGT 32T / 94T. Prop about 2,590 rpm at 7,600 engine rpm, tip speed about 745 ft/s.

QUIETER OPTION: Warp Drive 4-blade with HP blades, cut to 66 in ($1,680, about $405 more, plus about $30 for a larger pulley). It turns about 10% slower for the same power: about 2,340 rpm, tip speed about 675 ft/s. Thrust and top speed are about the same; it runs smoother and is noticeably quieter for the operator and from shore: about 3-4 dBA less prop noise at 50 ft (estimate). The total gain depends on the exhaust: with a loud exhaust it is only about 1-1.5 dBA, so fit a good muffler either way. Neither version is expected to meet the California shoreline limit (75 dB(A), SAE J1970): estimates are about 82-84 dBA (3-blade) and 79-82 dBA (4-blade) at 50 ft. Florida's airboat law (s. 327.391) requires a muffler but gives no decibel figure in the statute. The stock silencer is the muffler here.

IF YOU CHOOSE THE 4-BLADE, change these when you build:

1. BOM line 32: order the 4-blade (Warp Drive HP blades, 4-blade, up to 74 in, $1,680), cut to 66 in, with the same SAE #2 face plate and flange.
2. BOM line 31: order the large (prop shaft) pulley as 8MGT 104T instead of 94T. The small (crank PTO) pulley stays 32T.
3. Belt length: with the 52T pulley the belt is longer for the same center distance (about 30 in). Recalculate the belt length in Gates DesignFlex (or the pulley maker's calculator) before ordering, and check that the jack-bolt slots in the bearing plates still give tensioning travel both ways.
4. Clearance: the 104T pulley is about 1 in larger in diameter (10.43 in pitch diameter against 9.42 in): check its clearance to the stand, the engine and the cage.
5. Loads: prop torque rises about 11% (to about 270 lb-ft) and the turning (gyroscopic) load about 20%. The 1-1/2 in shaft and the bearings still have margin. To recheck, move the 104T belt to the top of the BELTS list in CAD/analysis/etec_reduction.js and run it.
6. Weight: about 3-4 lb more.
7. Pitch setting (both props): start at the prop maker's suggested pitch, do a tied-down full-throttle run, and adjust the pitch until the engine reaches about 7,600 rpm (never over its 7,900 rpm maximum). Never exceed the prop maker's maximum rpm (Warp Drive does not publish it: get it from them before ordering).

Everything else in the build (stand, cage, prop shaft, bearings, flange) is the same for both props.

Skills: Reading a spec sheet; comparing options

2.1 Verify the layout on your boat and make templates

Depends on: 1.1 | Alongside: 1.2 (if not yet done) | About 8 h DIY, 4 h pro

Goal: confirm the design fits your hull before cutting anything.

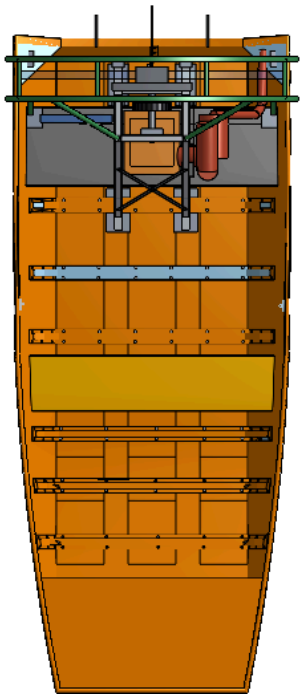
1. Measure from the transom and centerline: rib positions, rear bench (about X 13-33, top about Z 13.7), middle bench (about X 75.5-89.5, top about Z 12.5), gunwale height, transom height.
2. Check what the design depends on:

- Bench panels: their gauge, how they fasten to the hull, and whether they stop at about Z 2.4.
- Corner caps: whether they are aluminum and strong enough to carry a cage leg.
- Rear and middle benches: what is inside, and whether the access pocket and dry box opening are practical.
- Rib height at Y +/-8.5, and the hull bottom gauge.
- Transom condition around Z 4.5, where the clips bolt on.

- Front deck: strength under 45 lb of fuel, and how to fasten the straps.
- Rib height where the stick base and console foot land.

3. Weigh and measure the engine, tuned pipe, silencer, oil tank and radiator. Update the layout if they differ from the CAD envelopes, and choose the engine roll angle so the stock pipe fits. The engine sits about X 14-30.
4. Cut 1/4 in plywood templates for the four shoes, four cage foot pads, dry box opening, battery switch and bench sleeve hole.
5. Lay them out in the boat and write the measured numbers on them. Photograph everything.

If anything differs from the CAD by more than 1/2 in, stop and update the CAD before going on.



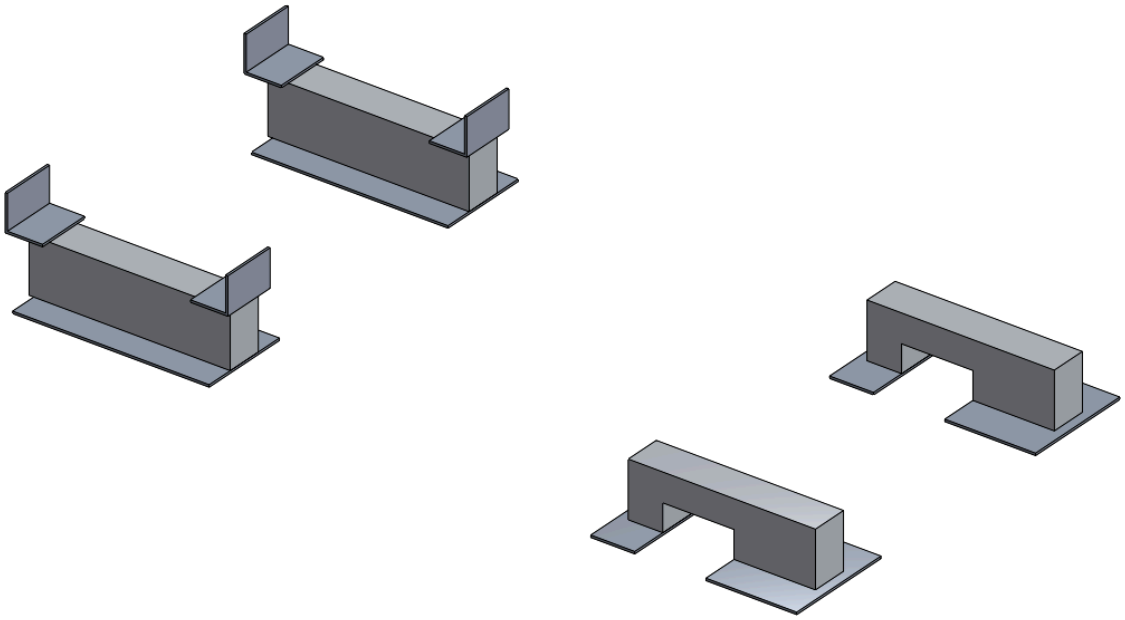
Skills: Measuring and layout; Reading the CAD model; Basic woodworking (templates)

2.2 Fabricate the shoes, doublers and clips (shop)

Depends on: 2.1 | Alongside: 2.3, 2.4, 2.5, 2.8 | About 6 h DIY, 3 h pro

Goal: four separate shoes, not continuous beds, so the rear bench and its flotation stay uncut.

1. Aft shoes (2): 2 x 3 x 1/8 in 6061 tube, X 2-12.75, at Y +/-8.5.
2. Forward shoes (2): the same tube, X 34-44, notched over the rib at X 36-39.5. Leave about 0.08 in over the rib top, so a hard landing bears on the rib.
3. Doublers, 1/8 in 5052: 5 x 10.5 in under each aft shoe. Under each forward shoe, two 6 in wide pieces, cut around the rib.
4. Clips: 2 x 2 x 1/8 in angle, 3 in wide. One at each end of each aft shoe, to the transom and to the rear bench's aft panel.
5. Drill for six 1/4-20 bolts per shoe. Cut a crush sleeve for each bolt, so the bolts can be tightened without collapsing the tube.
6. Deburr everything. Round all corners.



Skills: Cutting aluminum; Drilling; Layout from templates

2.3 Cut and drill the stand side frames (shop)

Depends on: 2.1 | Alongside: 2.2, 2.4, 2.5, 2.8 | About 8 h DIY, 4 h pro

Goal: all stand parts cut and drilled, ready to bolt together in 2.8. No welding is needed.

Side frames at Y +/-8.5, 6061, 1/8 in wall:

1. Aft posts: 2 x 2 in, X 9, from the shoe to Z 57.
2. Rails: 2 x 2 in, Z 14.9, from X 9 to about X 36. They clear the bench top by only 0.2 in; add a rubber strip if they touch under load.
3. Forward posts: 1-1/2 x 1-1/2 in, X 21, from the rail to Z 56.75.
4. Top rails: 1-1/2 x 1-1/2 in, Z 56, X 9-21.
5. Diagonals: 1-1/2 x 1-1/2 in, from the top of the forward post down to the forward shoe at X 40.
6. Ties across the boat: 1-1/2 x 1-1/2 in, at X 9 Z 14.9, X 9 Z 56 and X 21 Z 56.
7. Cross-straps: two 1 x 1/8 in flats in an X on the forward face of the diagonals, below the cylinders.
8. Gussets: 1/4 in 6061 at the joints, with four 5/16 in bolts on each side of each joint. Drill them as templates so both side frames match.

The rail is the most heavily loaded member, with a load factor of about 1.4 assuming engine mounts at X 17 and X 27. If your real mounts differ, recheck it with CAD/analysis/stand_800r.js.

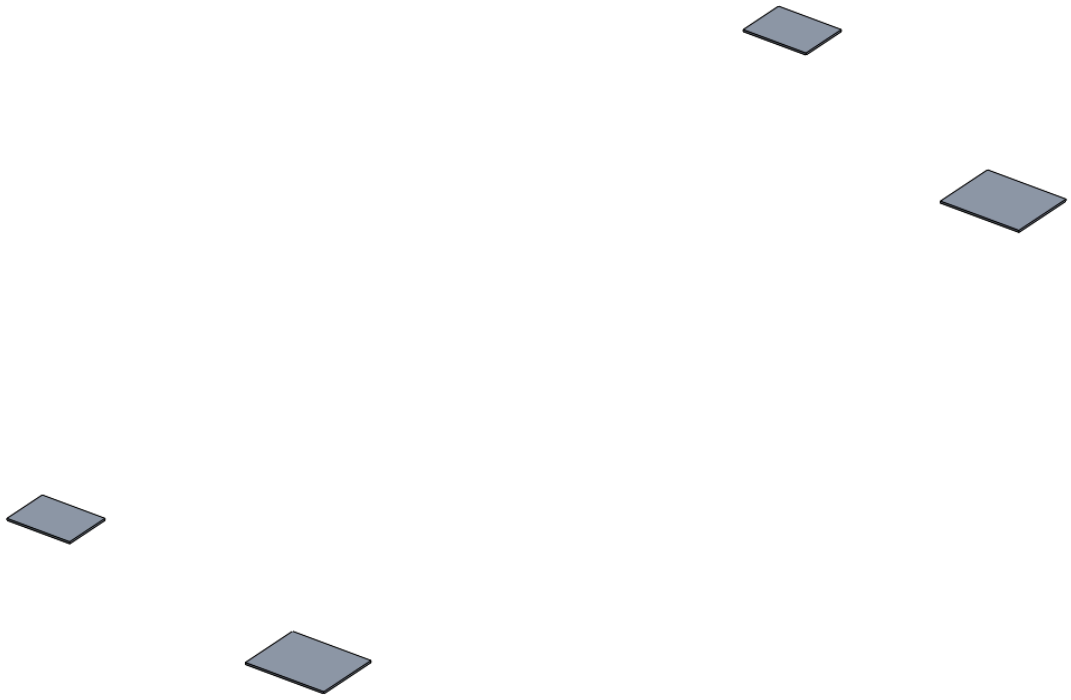
Skills: Cutting aluminum; Accurate drilling; Layout; Reading drawings

2.4 Fabricate cage foot pads and backing plates (shop)

Depends on: 2.1 | Alongside: 2.2, 2.3, 2.5, 2.8 | About 3 h DIY, 1.5 h pro

1. Aft pads (2): 1/8 in 6061, 4 x 3 in, for the transom corner caps, at about X -1 to 3 and Y +/-29.75 to 32.75.
2. Forward pads (2): 1/8 in 6061, 5 x 4 in, for the rear bench top, at about X 13.25-18.25 and Y +/-27.5 to 31.5.
3. A matching backing plate for each pad.

4. Drill four 1/4-20 holes through each pad and backing plate as a pair, and mark them so they stay matched.
5. Round the corners and deburr.



Skills: Cutting aluminum; Drilling; Layout

2.5 Open the benches for the dry box, switch and harness (boat)

Depends on: 2.1 | Alongside: 2.2, 2.3, 2.4, 2.8 | About 6 h DIY, 3 h pro

Goal: make the openings with as little foam loss as possible. Keep the removed foam volume in a notebook.

Middle bench:

1. Cut the forward panel opening for the dry box (about 10.6 x 9.7 x 6.9 in case, X 82.6-89.5, Z 2-11.7). Remove only the foam it needs. Fit the 1/8 in trim.
2. Cut the opening for the battery switch in the forward panel, port of the dry box, at about Y 8, Z 9.
3. Bore through the bench for the 2-1/4 x 0.065 in aluminum sleeve, from the box's aft face to the aft panel (X 82.6 to 75.3, on the centerline, Z 6 falling to Z 5). Deburr the sleeve ends and fit protective edge trim. Bed it so water can't reach the foam.

Rear bench:

4. Cut a small access pocket through the aft panel into the foam, so you can reach the forward cage pad backing plates. Refill and seal it after 2.7.
5. Seal all exposed foam. Restore any lost flotation volume elsewhere with closed-cell pour foam.

Skills: Cutting sheet aluminum; Drilling; Foam work; Marine sealant

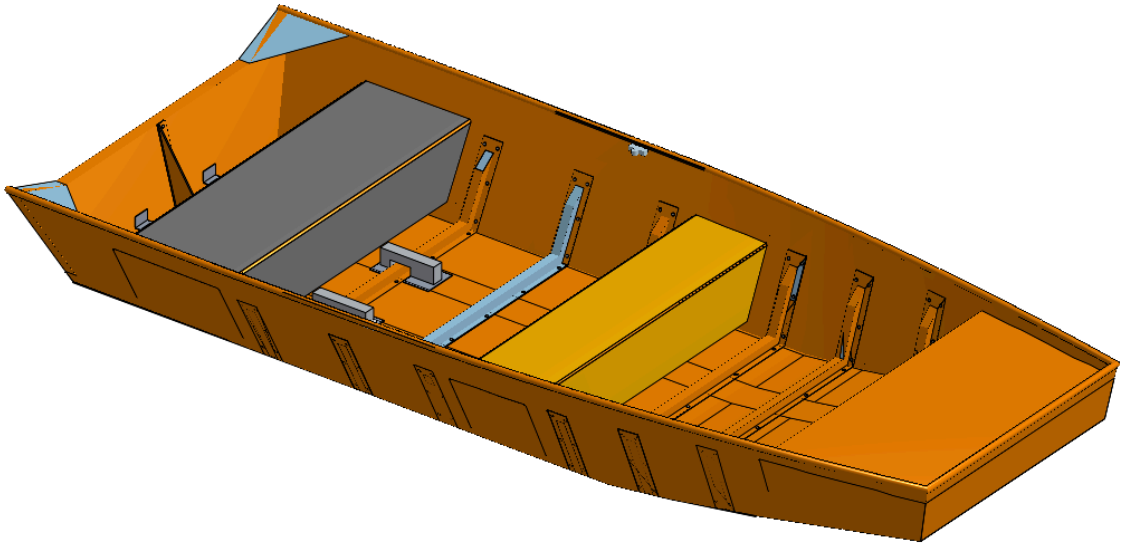
2.6 Install the shoes and doublers (boat)

Depends on: 2.2, 2.5 | Alongside: 2.4, 2.8 | About 8 h DIY, 4 h pro

Goal: four shoes tied to the bottom, the rib, the transom and the bench panel.

1. Dry-fit all four shoes on their doublers. Check they are parallel, at Y +/-8.5, and level across the boat.
2. Drill through shoe, doubler and bottom from inside. Fit the crush sleeves.

3. Bolt with six 1/4-20 stainless bolts per shoe and sealant in every hole. Use truss heads outside, under the 3/16 in UHMW thickness. The bolts pass between the centerline and chine strips.
4. Aft shoes: fit the clips to the transom (about Z 4.5) and to the rear bench's aft panel. These carry the uplift: about 394 lbf per side at full power, so don't skip them.
5. Forward shoes: check the rib notch. The shoe should nearly touch the rib top, with about 0.08 in of clearance.
6. Water-test the bottom and fix any weeps before going on.

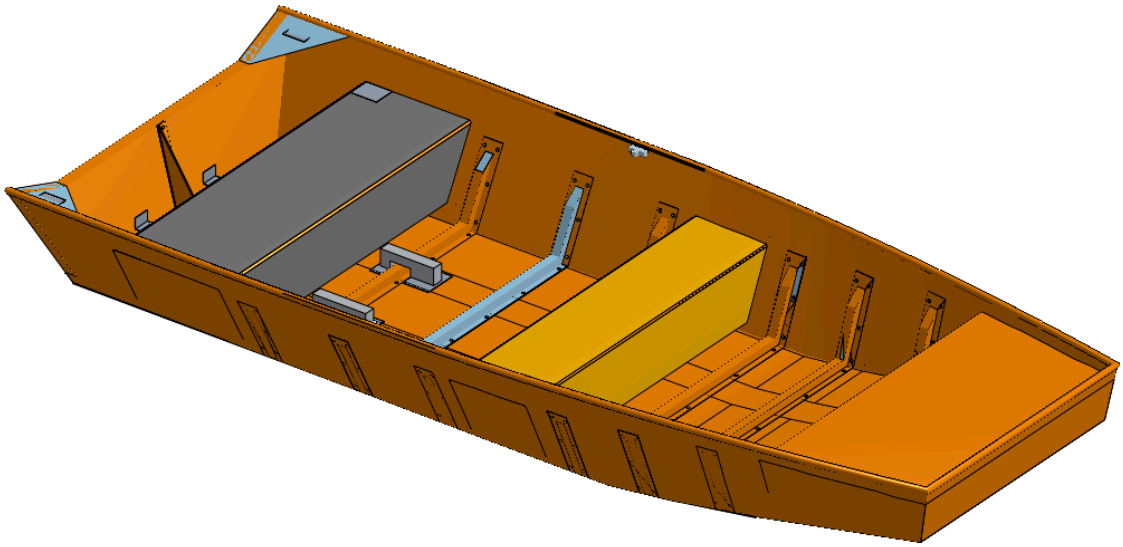


Skills: Drilling aluminum; Measuring and leveling; Bolting with a helper; Marine sealant

2.7 Install the cage foot pads (boat)

Depends on: 2.4, 2.5 | Alongside: 2.6, 2.8, 2.9 | About 4 h DIY, 2 h pro

1. Aft pads: bolt to the transom corner caps with backing plates and four 1/4-20 bolts each, at about Z 17.9.
2. Forward pads: bolt to the rear bench top next to the hull side, with backing plates reached through the access pocket from 2.5.
3. Check that the pads are level across the boat and at the right height for the cage legs.
4. Refill and seal the access pockets.
5. Water-test.



Skills: Drilling aluminum; Measuring and leveling; Bolting with a helper; Marine sealant

2.8 Build the stand and drivetrain module (shop)

Depends on: 2.1, 2.3 | Alongside: 2.2, 2.4-2.7 (biggest parallel task) | About 28 h DIY, 14 h pro

Goal: the whole drivetrain built and checked on the floor as one liftable unit.

Stand:

1. Use a plywood jig with the shoe spacing. Bolt up both side frames, ties, gussets and cross-straps.

Bearing plates (BOM line 31, cut by a waterjet or CNC shop):

2. Aft plate 3/8 in 6061 at X 8, forward plate 1/2 in 6061 at X 20. Each has two petals either side of the bearing, a 1/2 in rib, a 1 in border, and solid ends with the tensioning slots.
3. Round and polish every cut edge. Tool marks are where fatigue cracks start. Anodize if you want the finish.
4. Bolt the plates between the side frames, with jack bolts for belt tension.

Prop shaft:

5. 1-1/2 in 4140 shaft on a tapered-roller flange bearing forward (carries the thrust) and a UCF208-24 ball bearing aft. Fit two clamp collars either side of the forward bearing's inner ring. The set screws only locate it.
6. Fit the 94T pulley (104T for the 4-blade) at about X 13. Both pulleys must be flanged and dynamically balanced.

Engine:

7. Confirm the PTO taper and side load from the service manual before machining the 32T hub. If the manual can't confirm the side load, add an outboard support bearing.
8. Mount the engine on the sled rubber mounts, crank along X, PTO aft, crank at Z 22.72, 0.72 in higher than the original layout. That gives 29.28 in centers for the 8MGT-2000-36 belt.
9. Fit the belt. Align the pulleys with a straightedge. Tension with the jack bolts to the Gates value, then fit the belt guard. A wrong-tension belt fails early.

Attached to the stand:

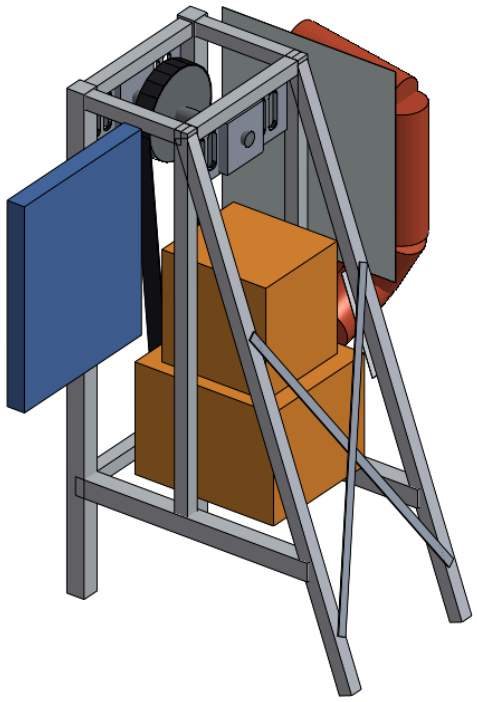
10. Oil tank, upright, on the starboard side frame, X 12.6-22.4, Z 17-26, filler on top. Keep the sled oil level sensor.
11. Radiator, 18 x 22 x 2 in, starboard, X 15-17, Y -28 to -10, Z 34-56, with the overflow bottle. It must see clean air, never air from behind the exhaust.
12. Tuned pipe and stock silencer on the port side, unmodified. Keep the sled springs and ball joint, and use a flexible joint between engine-mounted and stand-mounted parts.
13. Heat shields on standoffs between the exhaust and the stand, belt and bearings. Shields, not wrap.

Finish:

14. Add two lifting points at the balance point.
15. Torque every fastener and paint a torque stripe across each nut.
16. Weigh the module and write the weight on the frame. Update CAD/analysis/minimal_1448.js. The engine's 79 lb is unverified.

The prop is NOT fitted here (see 2.15).

Belt tension, new belt: 457 to 499 lb static, or 33 to 36 lb at mid-span to deflect the span 0.45 in. Set it with the jack bolts, keep the shafts parallel, and lock the jack bolts.



Skills: Mechanical assembly; Engine mechanics; Torque wrench use; Reading drawings; Belt tensioning

2.9 Rig the lift and do a trial lift (shop)

Depends on: 2.8 | Alongside: 2.7 | About 3 h DIY, 1.5 h pro

1. Set up an engine hoist or gantry rated for at least twice the module weight, with rated slings and shackles.
2. Lift 2 in off the floor and hold for 5 minutes. Check balance, that the stand does not rack, and that nothing touches the slings.
3. Plan the path into the boat: boat on its trailer, chocked, tongue on a stand, area clear.

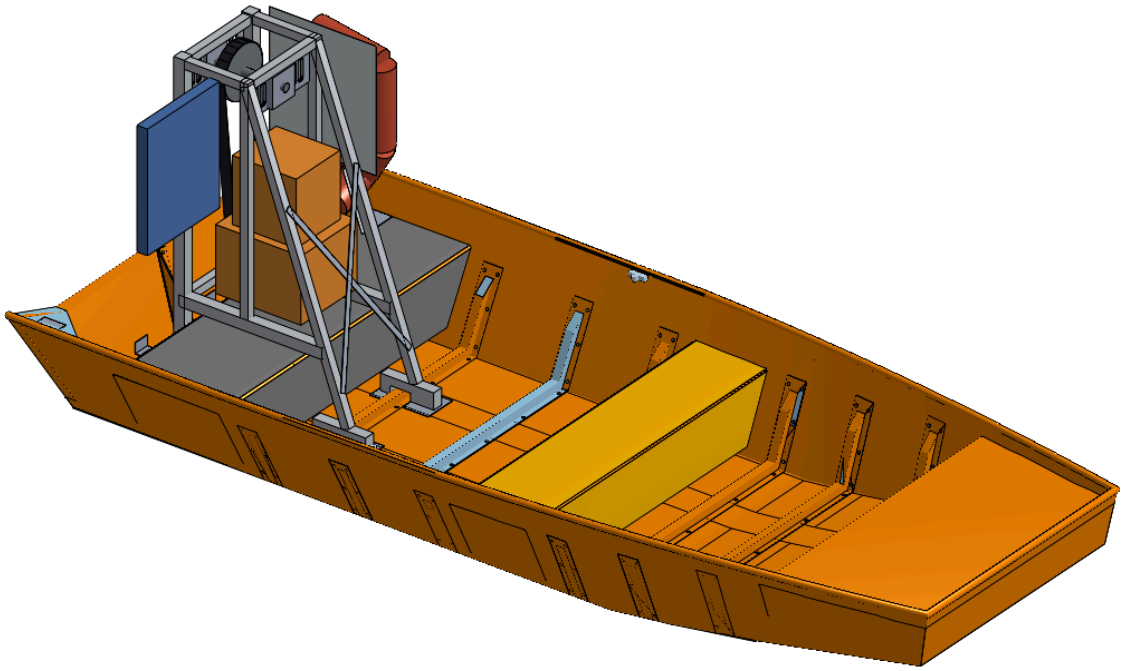
Never put any part of your body under a suspended load.

Skills: Rigging and safe lifting; Engine hoist operation

2.10 Lower the module into the boat and bolt it down (boat, 3 people)

Depends on: 2.6, 2.9 | Alongside: - | About 4 h DIY, 2 h pro

1. Lift and move slowly over the boat, with one person on the hoist and two guiding.
2. Lower the rails over the rear bench. They clear the bench top by only 0.2 in, so go slowly.
3. Land the aft posts on the aft shoes and the diagonals on the forward shoes. Line up the holes with drift pins.
4. Bolt down with threadlocker. Torque and stripe.
5. Check that nothing touches the benches, hull or ribs, and that the engine mounts aren't bottomed.
6. Check the module against the hull: engine to stand about 0.75 in, and the pipe's 4-6 in from the belt and bearings.



Skills: Rigging; Teamwork (3 people); Torque wrench use

2.11 Connect electrical, fuel, oil, cooling, exhaust and controls (boat)

Depends on: 2.10 | Alongside: 2.12, 2.13 | About 48 h DIY, 24 h pro

Do the wiring with the battery disconnected. Work from the schematic and wire list; each wire has a number, so label both ends.

Cockpit:

1. Dry box: seat the case in the middle bench with the lithium battery, fuse block and relays K1-K5.
2. Battery switch on the bench forward panel, port of the box.
3. Console: foot plate on the rib at X 108, post, and arm to the gunwale. Mount the pod, gauges, switches and phone holder. Drill the low oil LED between the temperature and exhaust gauges.
4. Fuel tank on the front deck step, X 134-152, strapped down, with the sled pump module and gauge sender.
5. Fire extinguisher next to the driver's right knee, about X 99, Y -26. Air horn in its clip.

Wiring:

6. Main power: W57 from the battery, with the 6 AWG fusible link, to the switch; W01 from the switch to the starter solenoid; W03 from the switch through MF1 to the hub; W02 and W04 for the negative side.
7. F7 and the bilge pump feed are inline at the battery, ahead of the switch, so the automatic pump always has power.
8. Harness aft: through the sleeve in the bench, then in the 2 in conduit on rib standoffs, centerline at Z 4. Support every 18 in. Use lined metal clamps near the engine, belt and prop and along the cage. Stow gear beside the conduit, not on it.
9. Harness forward: dash run to starboard, bow run to port. Keep wires 9 in from dry exhaust unless there is a thermal barrier, and never run them directly above it.
10. TB1 and TB2 are screw bars: one ring terminal on each screw.
11. Side lights: red to port, green to starboard, on the hull sides at X 108, Z 17. Fit a wedge so each stands vertical on the flared side, and use a sealed gland through the hull.
12. Low oil light: connect W60, W61 and W62. First confirm from the service manual that the sensor closes on low oil.
13. Engine side: J1 connector, DESS and lanyard kill, tach, temperature sender, exhaust probe. Confirm everything marked "confirm" on the schematic from the service manual.

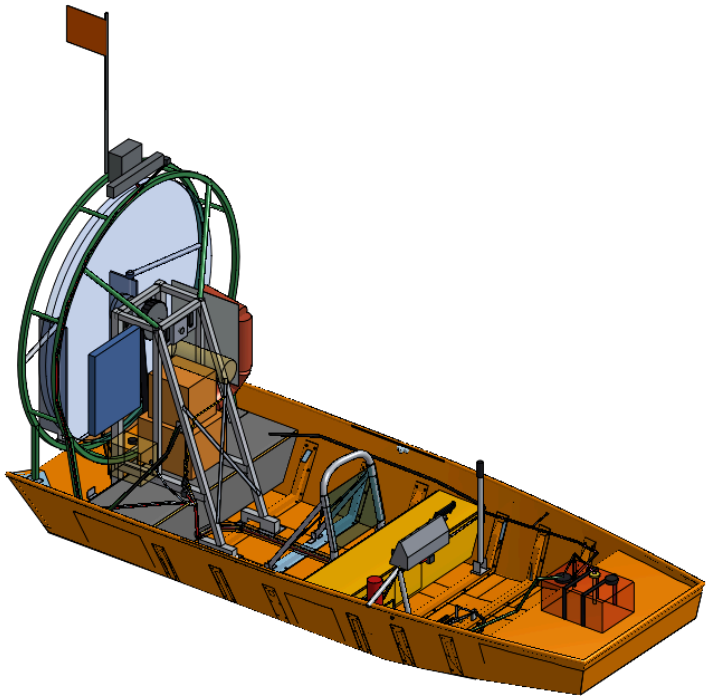
Engine systems:

14. Fuel line along the starboard side, about 1 in inboard of the skin, all joints clamped. Keep it away from battery cables.
15. Oil: connect the tank to the pump, fill it, and bleed the line exactly as the service manual says. This engine has no oil in the gas, so a bad bleed can destroy it. Run straight gasoline; never run the tank dry.
16. Cooling: connect the hoses and thermostat, fill and bleed. Trapped air here overheats the engine.
17. Tailpipe to the port transom corner: outlet at the transom top or higher, pointing aft and slightly outboard, never into the prop. Fit the pipe guard where a hand or knee could touch it.

Throttle:

18. Pedal, 3/16 in plate, at Y -2.5, hinged at the heel, with the rib at X 108 as its stop. Heel rests held by the UHMW strip screws.
19. Throttle cable from the post on the rib at X 108, along the starboard side to the engine, with a return spring at the pedal as well as at the throttle body. Confirm full open and full close. The engine must return to idle with your foot off.

Before moving on, check each circuit with a meter before fuses go in, then test the lights, gauges, bilge pump and low oil light. The starter stays disconnected until 2.14.



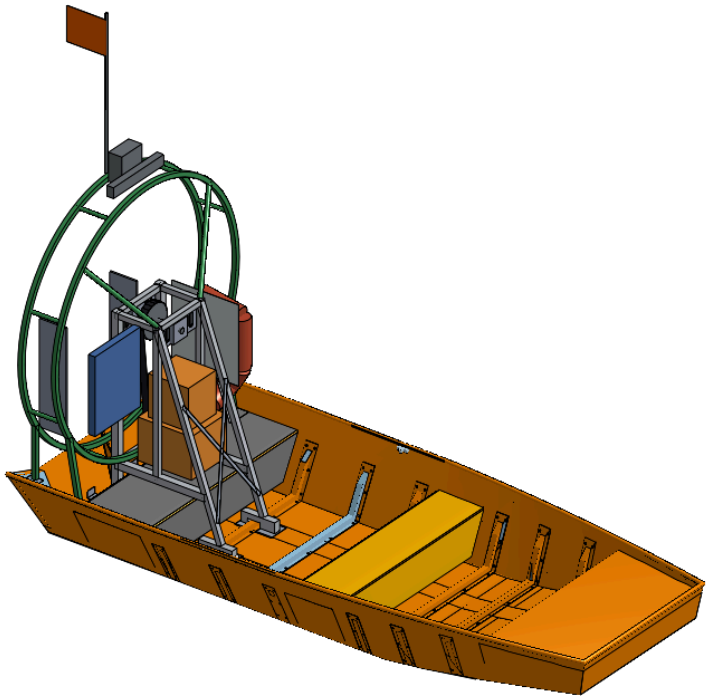
Skills: Marine wiring and crimping; Fuel system basics; Cooling system; Cable routing; Working from a schematic

2.12 Build the prop cage around the prop position (boat)

Depends on: 2.7, 2.10 | Alongside: 2.11, 2.13 | About 16 h DIY, 8 h pro

Goal: a cage built to the real prop position, with a way to fit the prop last.

1. Make a prop template: a 3/4 in plywood disc, 66 in diameter, bolted to the hub in place of the prop. Turn it by hand.
2. Build two hoops, 36 in radius, at X 1 and X 11, four legs at Y +/-30.5 on the pads from 2.7, and eight stringers.
3. Fit the two 1 x 0.065 in struts from the forward hoop to the top of the stand at X 21. Use rigid struts, never wires: a failed wire ahead of the disc can be drawn into the prop. Add rubber isolators at the strut ends if vibration calls for them.
4. Check clearance with the template: at least 2 in all round, and at least 2 in to the stand and bearing plates. The aft bearing corners and the aft hoop are tight, so check them against the real hub.
5. Make one section removable with bolts, big enough to pass the prop through. Label it.
6. Fit the mesh guard on the front and rim only, never in the prop wake: mesh there costs about 220 lbf of thrust. The cage is not a step. Remove the template.
7. Cage top: 20 in LED bar, searchlight and all-round white light. Connect their wires.
8. Flag: orange 10 x 12 in flag on the fold-down fiberglass whip, on the aft side of the cage, at least 10 ft above the bottom. Check the current FWC rule.

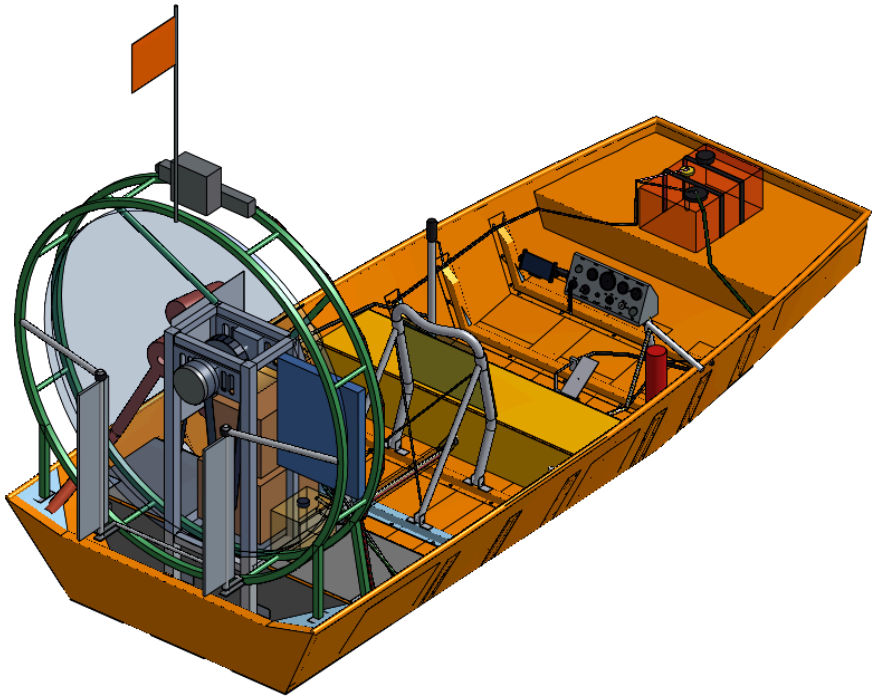


Skills: Mechanical assembly; Measuring; Tube bending or a bending shop; Bolting

2.13 Install the stick, rudders and steering linkage

Depends on: 2.12 | Alongside: 2.11 | About 8 h DIY, 4 h pro

1. Stick: base plate on the rib at X 95.25, Y 21, with the pivot cheeks. Steering stick, 1-1/2 x 0.120 in, and grip. Fit a stop on the cheeks that takes a hard pull before the rudder cable does.
2. Rudders: 8 x 30 in, hinged at 25% chord, on 1 x 0.125 in 6061 shafts. A 3/4 in shaft fails the load check, so don't downsize. Lower bracket on the aft hoop, upper arm out to the hoop.
3. Linkage: horn on each shaft, tie bar at X 0, and the 4.3 in outboard arm on the port shaft.
4. Cable: about 14 ft, along the port side, across the top of the rear bench to Y 18, under the hoops, to a clamp on the aft hoop.
5. Direction: stick forward turns left. Swap the cable to the other side of the rudder shaft to reverse it, and pick the habit before first running.
6. Set rudder stops at about 25 degrees each way, with equal travel and no binding. Check the tie bar at full rudder: it clears the hoop by only about 0.13 in at center.
7. Cotter pins or safety wire on every pivot.
8. Trim tab: fit the small tab and turnbuckle on the transom, to starboard (BOM line 80), with sealed bolts and a backing plate. Start it neutral.



Skills: Mechanical assembly; Linkage adjustment

2.14 Pre-test inspection and sign-off

Depends on: 2.11, 2.12, 2.13 | Alongside: - | About 2 h DIY, 1 h pro

1. Work through Part A of reports/Static_Test_Protocol_Drivetrain.md.
2. Re-torque all structural bolts and check every torque stripe.
3. Nothing loose inside the cage or the boat. Fluids full, no leaks. Belt tension checked.
4. Oil line bled, coolant bled, low oil light tested.
5. Connect the starter. Turn the engine by hand first to confirm the prop's rotation.
6. Two people sign the checklist.

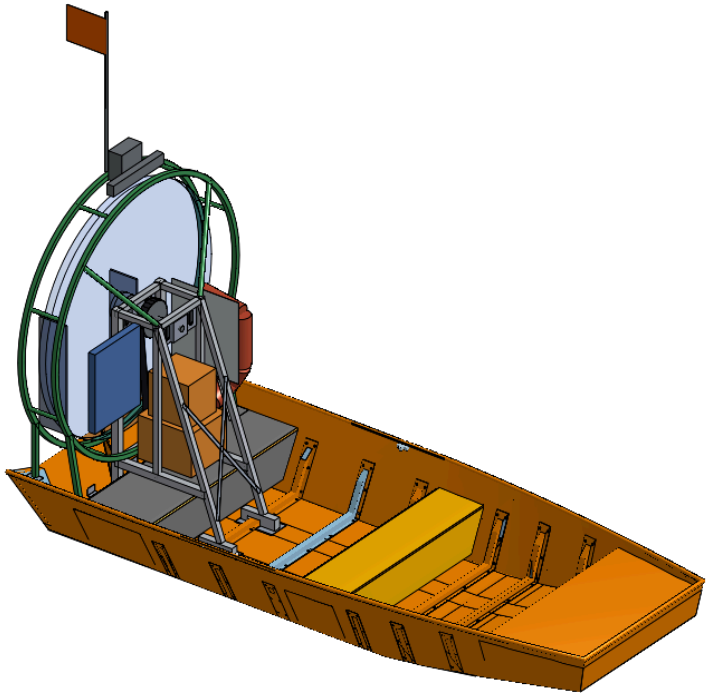
Skills: Inspection; Torque wrench use

2.15 Install the propeller (last step before testing)

Depends on: 2.14 | Alongside: - | About 3 h DIY, 1.5 h pro

1. Lock out: key out, lanyard off, battery switch off.
2. Remove the cage section from 2.12.
3. Inspect the prop for damage. Check static balance and blade track, within 1/16 in.
4. Mount it with the face plate and the maker's bolts. Warp Drive HP: 3/8 in mounting bolts to 30 ft-lb in 5 ft-lb steps; 1/4 in blade clamping bolts to 10 ft-lb in 2 ft-lb steps, torqued at the nuts. Star pattern. Torque stripe or safety wire.
5. Turn it by hand. Check at least 2 in of tip clearance all round, and that nothing touches.
6. Reinstall the cage section.
7. Set the blade pitch on a tied-down run so the engine reaches 7,400 to 7,500 rpm at full throttle, never over 7,900 engine rpm or 3,200 prop rpm. The maker recommends static rpm 400 to 500 under the engine maximum. Adjust in 1 degree steps, within 6 to 18 degrees.

8. Re-torque the prop bolts after the first run, then check them at 5 hours, at 50 hours, and at least yearly. Limit full throttle tied to the dock to about 1 minute at a time, and watch the temperature.



Skills: Mechanical assembly; Torque wrench use; Prop balancing and tracking

2.16 Build and fit the removable grab rail and sling

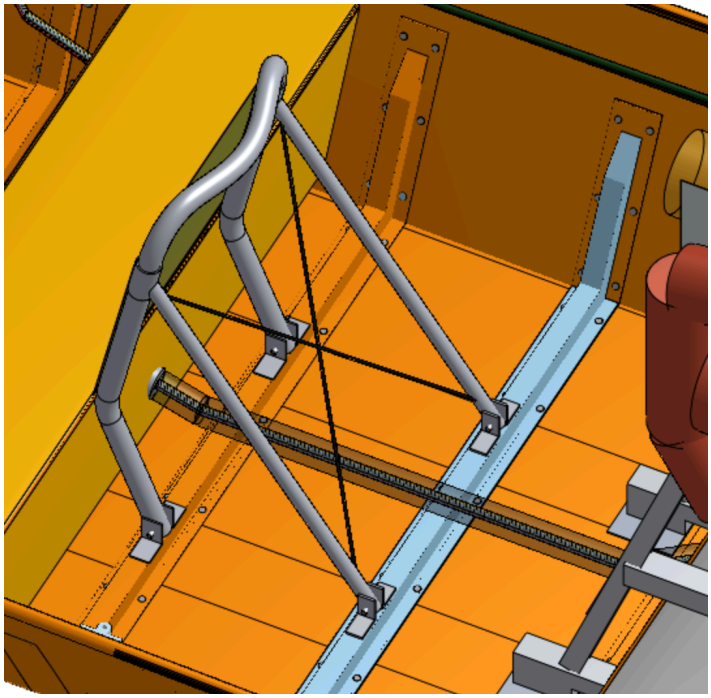
Depends on: 2.1 | Alongside: 2.11, 2.12, 2.13 | About 8 h DIY, 4 h pro

A removable hoop behind the driver that works as a grab rail, with a laced mesh sling as a backrest. Good for 300 lb in any direction. The hoop carries that load, not the fabric. It stands on two ribs, so nothing is welded to the hull and no hole goes through the bottom.

1. Check the ribs on your boat first: tops about 1.3 in wide at X 54.75 and 1.5 in wide at X 70.65, top at Z 1.62, sound at Y +/-9.25. Measure the top thickness. One stay foot can see about 480 lb of uplift. If yours differ, move the pins to suit and keep the bend at X 74.6, Z 12.5.
2. Try the lean first: sit on the bench against a board at 12 degrees and work the stick to full travel. If you can't reach it comfortably, reduce the lean before ordering the bends.
3. Hoop (shop): 1-1/2 x 0.083 in 6061-T6, about 7 ft, on a CNC mandrel bender. 20 in wide overall. Each leg runs forward from the foot pin to a 4 in radius bend at X 74.6, Z 12.5, then leans back 12 degrees to the top at Z 32, with 4 in corners and a 2 in aft bow across the top. All bends, no mitered welds: a welded joint at the leg bend would fail.
4. Stays (shop): 1 x 0.065 in 6061, about 29 in each, from 1-1/2 in below the corners down to the rib at X 54.75. Fish-mouth them into a 3 in long 1/8 in saddle around each leg. Tack on the boat with the pins in, then weld on the bench. Keep welds off the bends and the top bar.
5. Eye tabs: four, welded to the stays 1-1/2 in from each end, for the cables.
6. Pin holes: 5/16 in, reamed, each with a crush sleeve through the tube. A close fit keeps it from rattling.
7. Clips: a pair of 6061 angles at each mount, cheeks just clear of the tube. 2 x 2 x 1/8 in, 1-1/2 in wide at the feet; 2 x 2 x 3/16 in, 1-1/4 in wide at the stays, which take the side load. Fasten to the rib tops with sealed 1/4-20 aluminum rivnuts, two per clip.

8. Pins: 5/16 in stainless clevis pins with hitch clips. Tie each pin and its clip to a clip bracket with stainless lanyard wire so neither can get loose near the prop.
9. Cables: 1/8 in 7x19 stainless, crossed in the plane of the stays, from each top eye to the opposite bottom eye. Swaged, one turnbuckle each. Snug and equal, not bar tight, then wire the turnbuckles.
10. Sling: open PVC-coated polyester mesh, not solid fabric, so it doesn't catch prop air. About 26 x 16 in, double-hemmed, with grommets along the back edges and a bound slot at each stay. Wrap it around the upper legs from about Z 14 to Z 28 and lace it across the back with non-stretch cord. Keep it off the corners and top bar so they stay bare to grip. Pull it tight enough to sag no more than about 1-1/2 in under your back, and tie off the lacing.
11. Proof pull: 300 lb on the crane scale at the top center, forward, aft and each side. Nothing should move or stay bent. The hoop and cables take this, so check them, not the sling.

To carry bulky cargo: engine off, pull the four pins and lift the rail out with the sling and cables on. Leave it ashore or lash it down: a loose rail must not get near the prop. Refit all four pins and close the clips before the next run.



Skills: Measuring; Drilling; Rivnuts; Cable rigging; Basic sewing and grommets; Shop CNC bending and TIG welding (outsourced)

2.17 Fit the flotation foam

Depends on: 2.13 | Alongside: 2.16 | About 3 h DIY, 1.5 h pro

Foam so the boat floats level upside down, added to the foam already in the benches. Analysis: reports/Flotation_Inverted_v1.0.1.0.md. It relies on both benches being full of sound foam: check yours first.

1. Bench foam: keep it intact. Cut only what steps 2.5 and 2.6 need, and seal the cut faces.
2. Foam: closed-cell, 2 lb/ft3, fuel resistant: polyethylene, or XPS sealed in a fuel-proof skin. Not beadboard.
3. Bow wedge: 2 x 43.6 in, about 7.5 to 7.9 in deep, at the forward end of the space under the front shelf (X 154-156). Cut it to fit the hull bottom.
4. Aft corner blocks, two the same: 10 in long (X 3-13), from Y 10.5 out to the hull side, Z 2 to 10. About 14.4 in wide at the bottom and 18.4 in at the top. Cut them to fit the hull side.

5. Port block: put a 0.040 in aluminum shield between it and the tailpipe above, or trim it to keep 9 in from the pipe.
6. Strap all three in place with stainless or aluminum hardware. Keep them clear of the bilge pump and hose, and leave a drain path under them.
7. When the boat is finished, roll it in shallow water with a crane or helpers. It should float level with the hull bottom about 4 in clear. Trim the wedge to suit.

Skills: Measuring; Foam cutting; Mechanical assembly

Section 3 - Static Test

3.1 Static test of the installed drivetrain

Depends on: 2.15, 2.16, 2.17 | Alongside: - | About 6 h DIY, 4 h pro

Follow reports/Static_Test_Protocol_Drivetrain.md step by step, with the crew roles and PPE from reports/Static_Test_Plan_Drivetrain.md.

Record data on the protocol data sheets, then fill in reports/Static_Test_Performance_Report.md.

Do not run the boat on the water until the static test has passed and the report is signed.

Skills: Following a test procedure; Engine operation; Data recording; Safety officer role (crew of 4)